Week 5 · Lab 2

Version B

Graded Practice:

`?:`, `switch` and `while`

10 problems from LeetCode · 10 points each · checked by the TA at your desk

Week: 5 — second session (graded)
Duration: 2 hours 15 minutes
Format: individual problem solving in CLion
Covers: Lecture 4 — the ternary operator `?:`, `switch`/`case`, the `while` loop, `break`/`continue`

Name:

Group:

Date:

How this lab works

1. Solve the problems in CLion, one `.cpp` file per problem. Every program reads its input with `cin` and prints its answer with `cout`.
2. **When a problem works, raise your hand.** The TA comes to your desk, reads your code, runs a few inputs and writes your points in the box under the problem. Nothing is uploaded.
3. There are **10 problems, 10 points each** (100 in total): problems 1–3 are **easy**, 4–7 **medium** and 8–10 **hard**. Solve them in any order; a problem with no attempt earns 0 points. There is enough work for the whole session — keep going until the end.
4. Under each title, **Practises:** says which part of Lecture 4 the problem trains.
5. You may ask the TA for help — this is practice, not an exam. The TA may ask you a short question about your code; be ready to explain *why* it works.

All problems are from `leetcode.com`, restated for a program that reads from the keyboard (a list of numbers or a word is read one item at a time, because arrays and strings come later). On LeetCode itself the answer is written as a *function* inside `class Solution`; functions come next week.

Quick reference

Ternary operator `condition ? a : b` is a *value*. Put it in parentheses inside `cout`.

```
int mx = a > b ? a : b;
cout << (n % 2 == 0 ? "even" : "odd");
d = d < 0 ? -d : d;                  // |d|
```

`switch` for `int`/`char` values compared with constants. `break` leaves; without it execution falls through into the next `case`. Several labels may share one body.

```
switch (c) {
    case 'a': case 'e': case 'i':
    case 'o': case 'u': vowels++; break;
    case ' ':           spaces++; break;
    default:            others++;
}
```

`while`: initialise, test, update The test runs before every pass, so the body may run zero times.

```
int i = 1;                       // initialise
while (i <= n) {                 // test
    total += i;
    i++;                         // update
}
```

Reading *n* items without an array read one, use it, forget it:

```
int n;  cin >> n;
int best = -1000000000;
int i = 0;
while (i < n) {
    int x;  cin >> x;            // also: char c; cin >> c;
    if (x > best) { best = x; }  // running maximum
    i++;
}
```

A word given as `5 hello` is read the same way: `n`, then `n` single `char`s.

Digit by digit

```
while (n > 0) {
    int d = n % 10;              // last digit
    n /= 10;                     // drop it
}
```

`break` and `continue` `break` leaves the innermost loop (or `switch`) at once; `continue` jumps back to the test. In a `while`, do the update *before* a `continue`, or the loop never ends. A `while` inside a `while`: reset the inner loop's variables inside the outer loop.

Classic bugs a missing `break`; no update, or `while (...);` — an endless loop; `i != 10` with `i += 3`; `<` versus `<=`; a product that starts at 0; `int` overflow (use `long long`).

EASY

Problems 1–3 · 30 points

Short programs: one idea each. Test every branch, not just the examples.

1 Generate a String With Characters That Have Odd Counts 10 pts

LeetCode 1374 · `leetcode.com/problems/generate-a-string-with-characters-that-have-odd-counts`

Practises: a counting `while` loop and `?:`.

Read n ($1 \le n \le 500$) and print a word of n lowercase letters in which every letter that appears, appears an *odd* number of times. Use only `a` and `b`, with as many as as possible.

Input	Output	Why
4	aaab	3 times a, once b.
7	aaaaaaa	

TA: _____ / 10 pts initials _____

2 Palindrome Number 10 pts

LeetCode 9 · `leetcode.com/problems/palindrome-number`

Practises: a `while` loop over the digits.

Read an `int` x (negative values included). Print `true` if it reads the same backwards as forwards, otherwise `false`.

Input	Output	Why
121	true	
-121	false	Backwards it is 121-.
10	false	Backwards it is 01.

Hint: build the reversed number digit by digit and compare. Keep a copy of x — the loop destroys the number it works on. The reversed number may not fit in an `int`.

TA: _____ / 10 pts initials _____

3 Max Consecutive Ones 10 pts

LeetCode 485 · `leetcode.com/problems/max-consecutive-ones`

Practises: a `while` loop with `continue`, and a counter that resets.

Read n ($1 \le n \le 10^5$), then n numbers, each 0 or 1. Print the length of the longest run of consecutive 1s.

Input	Output	Why
6 1 1 0 1 1 1	3	
6 1 0 1 1 0 1	2	

Hint: when a 0 arrives, reset the run and `continue` *with the next number; otherwise extend the run.*

TA: _____ / 10 pts initials _____

MEDIUM

Problems 4–7 · 40 points

A few cases to get right. Test the boundaries.

4 Water Bottles 10 pts

LeetCode 1518 · `leetcode.com/problems/water-bottles`

Practises: a `while` loop that simulates, with `/` and `%`.

You have `numBottles` full bottles ($1 \le$ `numBottles` ≤ 100). Drinking one leaves an empty bottle, and `numExchange` empty bottles ($2 \le$ `numExchange` ≤ 100) can be exchanged for one full bottle. Read both numbers and print the largest number of bottles you can drink.

Input	Output	Why
`9 3`	`13`	Drink 9; 9 empties → 3 full; drink 3; 3 empties → 1; drink 1.
`15 4`	`19`	

Hint: keep the number of empty bottles. While you can exchange, exchange as many as possible; the new full bottles are drunk, and become empties too.

TA: ____ / 10 pts initials _____

5 Maximum Number of Balloons 10 pts

LeetCode 1189 · `leetcode.com/problems/maximum-number-of-balloons`

Practises: `switch` inside a loop over characters.

Read n ($1 \le n \le 10^4$), then a text of n lowercase letters (no spaces). Each letter can be used once. Print how many times the word `balloon` can be formed.

Input	Output	Why
`10 nlaebolko`	`1`	
`16 loonbalxballpoon`	`2`	
`8 leetcode`	`0`	

Hint: count `b`, `a`, `l`, `o`, `n` *with a* `switch`. *One* `balloon` *needs two* `l`*s and two* `o`*s.*

TA: ____ / 10 pts initials _____

6 Factorial Trailing Zeroes 10 pts

LeetCode 172 · `leetcode.com/problems/factorial-trailing-zeroes`

Practises: a `while` loop that steps by multiplying.

Read n ($0 \le n \le 10^4$) and print how many zeros $n! = 1 \cdot 2 \cdot \ldots \cdot n$ ends with.

Input	Output	Why
`3`	`0`	3! = 6.
`5`	`1`	5! = 120.
`0`	`0`	0! = 1.

Hint: $n!$ is far too big to compute (already 21! overflows `long` `long`*). Each trailing zero is a factor $10 = 2 \cdot 5$, and there are always more 2s than 5s. So count the 5s: every multiple of 5 gives one, every multiple of 25 one more, . . .*

TA: ____ / 10 pts initials _____

7 Longest Continuous Increasing Subsequence 10 pts

LeetCode 674 · `leetcode.com/problems/longest-continuous-increasing-subsequence`

Practises: a `while` loop that remembers the previous number.

Read n ($1 \le n \le 10^4$), then n integers. Print the length of the longest block of consecutive numbers that is *strictly* increasing.

Input	Output	Why
5 1 3 5 4 7	3	1 3 5.
5 2 2 2 2 2	1	Equal numbers do not increase.

TA: ____ / 10 pts initials _____

HARD

Problems 8–10 · 30 points

More cases and more care. Read the hint before you start.

8 Pascal's Triangle 10 pts

LeetCode 118 · `leetcode.com/problems/pascals-triangle`

Practises: a `while` loop inside a `while` loop.

Read `numRows` ($1 \le$ `numRows` ≤ 30) and print the first `numRows` rows of Pascal's triangle, one row per line, numbers separated by single spaces. Each number is the sum of the two numbers above it; every row starts and ends with 1.

Input	Output	Why
1	1	

Input 5 prints:

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
```

Hint: without arrays you cannot keep the row above — compute each number directly. In row r (counting from 0) the numbers are $C(r,0), C(r,1), \ldots, C(r,r)$, and $C(r,k+1) = C(r,k) \cdot (r-k)/(k+1)$, starting from $C(r,0) = 1$. Multiply before dividing, in a `long long`.

TA: ____ / 10 pts initials ____

9 Robot Bounded In Circle 10 pts

LeetCode 1041 · `leetcode.com/problems/robot-bounded-in-circle`

Practises: `switch` inside a loop, and a direction kept as a number.

A robot stands at $(0, 0)$ facing north. Read n ($1 \le n \le 100$), then n instructions written together: `G` = go one step forward, `L` = turn 90° left, `R` = turn 90° right. The robot repeats the instructions for ever. Print `true` if it never leaves some circle around the start, otherwise `false`.

Input	Output	Why
6 GGLLGG	true	It walks 2 north, turns round, walks back: home.
2 GG	false	It walks north for ever.
2 GL	true	It walks a small square.

Hint: simulate the instructions once. *Keep the direction as 0 = north, 1 = east, 2 = south, 3 = west; a right turn is* `(dir + 1) % 4`. *Afterwards the robot is bounded if it is back at $(0, 0)$ — or if it no longer faces north (then four rounds bring it home).*

TA: ____ / 10 pts initials ____

10 Count Primes 10 pts

LeetCode 204 · `leetcode.com/problems/count-primes`

Practises: nested loops with `break`.

Read n ($0 \le n \le 2 \cdot 10^5$) and print how many prime numbers are **strictly less** than n. (LeetCode allows n up to $5 \cdot 10^6$, which needs an array; here n is smaller.)

Input	Output	Why
`10`	`4`	2, 3, 5, 7.
`0`	`0`	
`1`	`0`	

Hint: test each i from 2 to $n-1$ with divisors $d = 2, 3, \ldots$ while $d \cdot d \le i$ — a divisor larger than $\sqrt{i}$ would have a partner smaller than $\sqrt{i}$. Leave the inner loop with `break` *as soon as a divisor is found.*

TA: ____ / 10 pts initials _____

Score sheet — Version B

	Problem	Level	Points	Earned	TA
1	Generate a String With Characters That Have Odd Counts (LC 1374)	Easy	10		
2	Palindrome Number (LC 9)	Easy	10		
3	Max Consecutive Ones (LC 485)	Easy	10		
4	Water Bottles (LC 1518)	Medium	10		
5	Maximum Number of Balloons (LC 1189)	Medium	10		
6	Factorial Trailing Zeroes (LC 172)	Medium	10		
7	Longest Continuous Increasing Subsequence (LC 674)	Medium	10		
8	Pascal's Triangle (LC 118)	Hard	10		
9	Robot Bounded In Circle (LC 1041)	Hard	10		
10	Count Primes (LC 204)	Hard	10		
	Total		**100**		

After the lab

Every problem here is on LeetCode under the number shown. Here the input is read one item at a time; on LeetCode it arrives as an array or a string. Once functions are covered, rewrite your solutions as a LeetCode `class Solution` and submit them.

End of Week 5 · Lab 2 · Version B